

Data Analyst vs Data Scientist vs AI Engineer

Steps involved in the **AI/ML Pipeline**

- Step1 : Data Collection
- Step2 : Data Cleaning & Preprocessing
- Step3 : Exploratory Data Analysis (EDA)
- Step4 : Feature Engineering
- Step5 : Model Selection & Training
- Step6 : Model Evaluation
- Step7 : Model Deployment

Skills for different Career Paths

1. Data Analyst

- Data Collection
 - SQL
 - Excel / Google Sheets
- Data Cleaning & Preprocessing
 - Python
 - Statistics
- EDA
 - Power BI
 - Tableau

2. Data Scientist

- Data Collection
 - SQL

- Data Cleaning & Preprocessing
 - Advanced Python
 - NumPy & Pandas
- EDA
 - Matplotlib
 - Seaborn
- Feature Engineering
- Model Selection & Training
 - Scikit-learn
 - Supervised ML
 - Unsupervised ML
 - Reinforcement Learning
 - Git & Github (additional)
 - Math
 - Statistics
 - Linear Algebra
 - Probability
 - Calculus
- Model Evaluation

3. AI Engineer

- Data Collection
 - SQL
- Data Cleaning & Preprocessing
 - Advanced Python
 - NumPy & Pandas
- EDA

- Matplotlib
 - Seaborn
- Feature Engineering
- Model Selection & Training
 - Scikit-learn
 - Supervised ML
 - Unsupervised ML
 - Reinforcement Learning
 - Git & Github (additional)
 - Math
 - Statistics
 - Linear Algebra
 - Probability
 - Calculus
 - Deep Learning (PyTorch or TensorFlow)
 - ANN
 - FNN
 - RNN
 - CNN
 - Transformer etc.
- Model Evaluation
- Model Deployment
 - Basic Deployment
 - API Building
 - DSA (optional, asked by some companies)

As of May 2026, the most relevant AI/ML projects focus on practical, production-ready integrations rather than just model training. Top industry demands emphasize Retrieval-Augmented Generation (RAG) pipelines, Multi-Agent systems, and MLOps. Depending on your skill level, consider exploring the following trending AI/ML projects:

- Advanced / Industry-Grade Projects (Production-Ready)**
 - Agentic Claims Processing System:** Build an AI agent using frameworks like LangGraph or CrewAI to process insurance or medical claims. It should parse documents, cross-reference databases, and generate structured decisions.
 - Multimodal RAG Chatbot:** Develop a question-answering system powered by vector databases that can ingest and reason over PDFs, images, and audio files.
 - Real-Time Model Observability Dashboard:** Combine tools like Kafka, Spark, and an MLOps platform to build a pipeline that monitors data drift and latency for deployed ML models.
 - Text-to-SQL Analytics Assistant:** Create a conversational tool that translates natural language queries into complex SQL queries to generate real-time business insights from databases.
- Intermediate Projects (Deep Learning & NLP)**
 - Custom LLM Fine-Tuning:** Take an open-source model and fine-tune it using PEFT and LoRA techniques on a highly specific domain dataset.
 - Credit Card Fraud Detection:** Utilize Scikit-learn, XGBoost, and anomaly detection algorithms to identify fraudulent transactions in real-time.
 - Disease Prediction with CNNs:** Train Convolutional Neural Networks on medical imaging datasets (like chest X-rays) to flag early stages of conditions like pneumonia.
 - AI News Summarizer:** Scrape live news articles and use NLP to categorize, analyze sentiment, and provide personalized news summaries.
- Beginner Projects (Foundational ML)**
 - Spam Email Classifier:** Build a classic text classification project using Python libraries like Scikit-learn and NLTK to filter out spam.
 - House Price or Customer Churn Prediction:** Use structured datasets from Kaggle to learn supervised machine learning tasks, such as linear regression or classification models.
 - Handwritten Digit Recognition:** Construct a foundational neural network (using TensorFlow or Keras) to classify numbers from image datasets.

4. Project Resources & Tutorials To get hands-on experience and access to guided source code for these modern project ideas, explore the following resources:

- For all levels:** Browse the [GeeksforGeeks AI Projects](#) for code guides across various difficulties.
- For structured growth:** Review the [Dataquest Machine Learning Projects](#) for step-by-step skill-building.
- For modern production engineering:** Check out the [Codebasics AI Projects](#) tailored to modern architectures.
- For applied ML:** Reference the [Digicrome Machine Learning Projects](#) for high-impact use cases in healthcare and finance.

Top AI/ML Projects on AWS & Azure Cloud in 2026 – K21 Academy
 6 Oct 2025 — [Azure AI/ML Projects in 2026](#) : * Chatbot Using Azure AI Search and OpenAI With Our Own Data. * Synthetic Data Generation with LLM.
 K21 Academy
 Five AI Projects for 2026
 18 Mar 2026 — [for your AI. career you need projects that are inspired by production grade AI. systems](#)
 systems that uses rag multimodel inputs ag...
 19:42
 YouTube-codebasics10